Lattice Enthalpy and Born–Haber Cycles

Building ionic solids from their elements

Name _____ Class _____ Date _____

BORN–HABER DATA (KJ/MOL)

LiF: ΔHf −616.0 · sublimation Li +159.3 · IE₁ Li +520.2 · atomization ½F₂ +79.4 · EA₁ F −328.2

KBr: ΔHf −393.8 · sublimation K +89.0 · IE₁ K +418.8 · ½Br₂(l) → Br(g) +111.9 · lattice −689.0

CaO: ΔHf −634.9 · sublimation Ca +177.8 · IE₁ Ca +589.8 · IE₂ Ca +1145.4 · atomization ½O₂ +249.2 · EA₁ O −141.0 · lattice −3414

MgCl₂: ΔHf −641.3 · sublimation Mg +147.1 · IE₁ Mg +737.7 · atomization ½Cl₂ +121.3 · EA₁ Cl −348.6

ΔH(latt) = ΔHf − (sublimation + atomization + ionization energies + electron affinities). Show units in every step.

A. Definitions

1. (a) Define lattice enthalpy. (b) Write equations, with state symbols, for the lattice enthalpy and for the enthalpy of formation of potassium bromide.

2. State whether each step is endothermic or exothermic: (a) $\mathrm{Ca(s)} \longrightarrow \mathrm{Ca(g)}$ (b) $\mathrm{O(g)} + \mathrm{e^-} \longrightarrow \mathrm{O^-(g)}$ (c) $\mathrm{O^-(g)} + \mathrm{e^-} \longrightarrow \mathrm{O^{2-}(g)}$ (d) $\mathrm{Ca(g)} \longrightarrow \mathrm{Ca^+(g)} + \mathrm{e^-}$ (e) $\mathrm{Ca^{2+}(g)} + \mathrm{O^{2-}(g)} \longrightarrow \mathrm{CaO(s)}$

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B. Calculations

3. Calculate the lattice enthalpy of lithium fluoride.

4. Use the KBr data to calculate the first electron affinity of bromine.

C. Trends

5. Put KCl, CaO, NaCl and MgO in order of increasingly negative lattice enthalpy. Explain your order.

6. (a) Use the CaO data to calculate the second electron affinity of oxygen. (b) Explain why it has this sign.

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D. Challenge

7. The lattice enthalpy of the hypothetical compound MgCl (containing $\mathrm{Mg^{+}}$) is estimated to be −753.0 kJ/mol. (a) Calculate $\Delta H_\mathrm{f}^\circ$ for MgCl. (b) Calculate ΔH for $2\,\mathrm{MgCl(s)} \longrightarrow \mathrm{Mg(s)} + \mathrm{MgCl_2(s)}$. (c) Explain why magnesium chloride is $\mathrm{MgCl_2}$, not MgCl.

8. For NaCl, the Born–Haber lattice enthalpy and the value calculated from a purely ionic model agree within about 2 %. For silver iodide, the Born–Haber value is much more negative than the model predicts. Explain both observations.

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